BIOT 6900 · Module 4 Assignment

Structure-Based Druggability of SAMHD1 and APOE in Alzheimer's Disease

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Targets. From my Module 2 AD list (targets_ad.csv) I took the two targets that my Module 3 report sent forward to structure: **SAMHD1** (Q9Y3Z3, 626 residues, Module 2 rank 1) and **APOE** (P02649, 317 residues, rank 4). SAMHD1 had the strongest disease evidence and the highest small-molecule prior (p_sm 0.824); APOE is the strongest genetic risk factor for late-onset AD, yet its sequence prior was near zero (p_sm 0.002). Experimental structures: **4BZC** chain A for SAMHD1 (wild-type catalytic core, residues 113–626, X-ray 2.88 Å, tetramer with dGTP-α-S bound) and **2L7B** chain A for APOE (full-length monomeric apoE3, NMR, model 1). Pockets were found with fpocket (run under WSL on Windows) and filtered with the course confidence rules.

Target	Mean pLDDT	% >90	% <50	Disordered stretches	Domains (PAE<8 Å)	Inter-domain PAE	RMSD (confident)
SAMHD1	88.2	74.4	9.6	1–36, 602–624	1 (38–600)	n/a	0.67 Å (472 pairs, 4BZC)
APOE	75.5	31.9	16.1	1–29, 31–40	2 (42–208; 226–303)	25.8 Å	32.35 Å overall (230 pairs, 2L7B)

Table 1. Confidence and validation summary (from structure_assessment_ad.csv and the notebook).

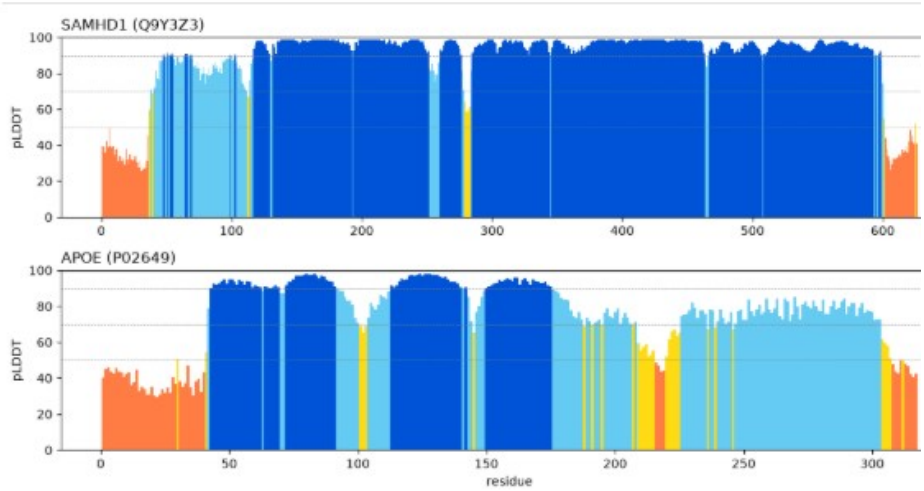


Figure 1. AlphaFold pLDDT along each sequence (dark blue >90, light blue 70–90, yellow 50–70, orange <50).

1. Which parts can I trust, and which can't I?

SAMHD1 is mostly very confident. The two disordered stretches match its biology: residues 1–36 are the N-terminal tail that carries the nuclear localisation signal, and 602–624 is the flexible C-terminal tail, which is also missing from the 4BZC crystal. A short dip near residue 280 matches a loop (278–283) that is also unresolved in the SAMHD1 crystal structures, so low pLDDT and missing density agree. The domain finder returned a single domain (38–600), but the PAE map (Figure 2) shows the SAM domain (about 40–110) as a separate dark block with a paler border to the rest. I would trust the SAM domain's shape but not its exact placement. Using the slide 12 decision table, I trust the **HD catalytic domain (about 115–600)** for pocket analysis: high pLDDT and dark PAE throughout.

APOE has two confident regions with uncertain arrangement. Residues 1–18 are the signal peptide (cleaved in the mature protein) and 19–40 the flexible N-terminus, both very low confidence. The **N-terminal four-helix bundle** (UniProt 42–182, mature 24–164, which contains the receptor-binding region) is very confident. The hinge and the **C-terminal lipid-binding domain** (226–303) are only 70–80, and the mean PAE between the two domains is **25.8 Å**, meaning their relative position is close to a guess. This fits APOE biology: the C-terminal domain binds lipid and changes shape when lipidated, and the way the two domains

contact each other differs between ε3 and ε4. I trust the bundle on its own for pockets, the C-terminal domain only cautiously, and nothing that depends on how the two domains sit together.

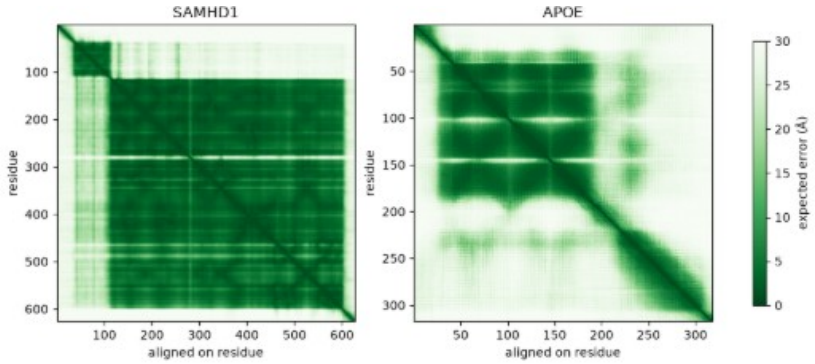


Figure 2. PAE heatmaps. Dark = confident relative position. APOE shows a dark bundle block and a pale block between the bundle and the C-terminal domain.

2. How well does each prediction match experiment?

SAMHD1 matches very closely. Pairing residues by sequence alignment (4BZC uses UniProt numbering) and superimposing 472 identical, confident Cα pairs gave **RMSD 0.67 Å**, well inside the 1–2 Å 'very close' range. The largest deviations are a few residues in a short surface loop (465: 6.5 Å, 464: 4.9 Å, 472–473: 2.8 Å) and residue 284 (2.7 Å), next to the 278–283 loop the crystal does not resolve (Figure 3). These are most likely **differences in context, not prediction errors**. The crystal is a dGTP-induced tetramer, and dGTP binding is known to rearrange the protein around the substrate pocket, whereas AlphaFold predicted a lone monomer with no nucleotides or metals. Flexible surface loops are also where crystal packing and the missing partners show up first.

APOE shows why per-domain comparison matters. The overall RMSD of **32.35 Å** is not meaningful, because one rigid fit cannot superimpose two domains whose arrangement AlphaFold was unsure about. This is the same lesson as the toy model in Part 1. Per domain, the finder's domain 1 (42–208) still gave 13.2 Å because it includes the hinge. On the **four-helix bundle alone (42–182) the RMSD is 2.60 Å** over 137 residues, so the bundle is correct. The numbering offset between 2L7B and UniProt was a constant 18 for all 299 aligned pairs, confirming the pairing. The C-terminal domain stays far apart even on its own (21.5 Å over 71 residues). Part of this is a real limit of the prediction (pLDDT only 70–80), and part is the experiment: 2L7B is an engineered monomeric mutant with five substitutions in the C-terminal domain, solved lipid-free in solution, a state in which that domain is known to be dynamic. Neither structure is likely to show the lipid-bound C-terminal domain that matters in the brain.

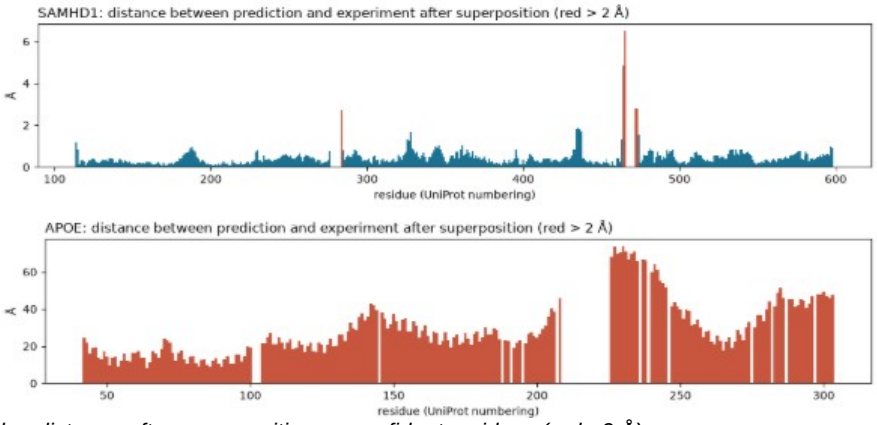


Figure 3. Per-residue distance after superposition on confident residues (red >2 Å).

3. Druggability calls

Target	Pocket (rank)	Score	Volume (Å³)	Mean pLDDT	PAE within	Verdict	Next best kept score
SAMHD1	3 (1)	0.483	1227	96.6	1.9 Å	keep	0.026 (pocket 33)
APOE	23 (1)	0.465	884	76.5	4.2 Å	keep	0.005 (pockets 2, 7)

Table 2. Best kept pocket per target. Full top-five tables are in the notebook (Step 6).

SAMHD1: "Structure supports a small-molecule pocket." Pocket 3 is the top-ranked pocket overall, sits in the most confident part of the model (pLDDT 96.6, PAE 1.9 Å) and is large (1,227 Å³). Its score (0.483) is far above every other pocket (≤0.026). Most importantly, the ligand-site check shows it covers **100% of the residues around the catalytic-site nucleotide** in 4BZC (Question 6). The prediction is also confirmed by experiment at 0.67 Å. This meets every condition in the decision table: a kept, well-scoring pocket in a confident region, confirmed by an experimental ligand.

APOE: "Possible but uncertain." fpocket found one well-scoring kept pocket (pocket 23, score 0.465, 884 Å³), so the protein is not pocket-less. Its lining residues are 31–56 and 163–174 (mature 13–38 and 145–156). That places it at one end of the four-helix bundle, between the start of helix 1 and helix 4, next to the receptor-binding region (mature 136–150). The bundle itself is trustworthy (2.60 Å vs 2L7B), but part of the pocket is closed off by residues 31–40, which lie in the low-confidence N-terminal stretch. This is why the pocket passes the rules only narrowly (mean pLDDT 76.5), and without that flexible segment the cavity may be shallower than fpocket suggests. There is no small-molecule ligand in 2L7B to check it against, and the overall domain arrangement is uncertain. I would not call it a 'no pocket' target, but I would not commit to it without an independent check. Because APOE is secreted, an **antibody** or an approach that changes APOE levels or lipidation remains a strong alternative, consistent with its higher antibody than small-molecule prior in Module 3 (0.016 vs 0.002).

4. Comparison with Module 2 and Module 3

For **SAMHD1** all three modules agree: rank 1 in Module 2, the top small-molecule prior in Module 3 (0.824), and now a confident, experimentally confirmed pocket. For **APOE** structure partly overturns the sequence prior. Module 3 scored it near zero because few apolipoproteins have drugs, and I suggested that might be a blind spot. Structure finds a kept pocket the sequence model could not see, which supports that idea, but the pocket is not yet validated. I would keep both targets in the pipeline, with different priorities. SAMHD1 is the stronger structural case, but it has a biological caveat. Its Module 2 signal was strong RNA down-regulation (log2FC −3.65), and loss of SAMHD1 causes Aicardi–Goutières syndrome, an interferon-driven brain disease. So the right drug may be an activator or a stabiliser rather than an inhibitor, and that question must be answered before chemistry starts. APOE stays as a high-value, high-uncertainty target that needs the follow-up below.

5. Limits that apply to my targets

Missing partners and oligomeric state. SAMHD1 works as a dGTP-induced tetramer, and its allosteric sites are formed between subunits, so a monomer prediction cannot show them properly (Question 6). APOE works on lipoprotein particles, and its lipid-bound form is not predicted. To address this, I would model the SAMHD1 tetramer with nucleotides (AlphaFold-Multimer or Boltz-2 on the cluster) and compare APOE with structures or models of lipidated apoE.

One static snapshot, with no metals or mutant forms. AlphaFold gives one wild-type conformation. SAMHD1's active site holds metal ions (Mn and Mg in 4BZC) that are absent from the model, and its substrate pocket changes shape on dGTP binding. For APOE, the AD-relevant form is ε4 (C130R, mature C112R), whose domain interaction differs from the ε3 sequence modelled here. To address this, I would model APOE ε4 alongside ε3, compare with ESMFold, and test whether pocket 23 persists using short molecular dynamics or fragment screening.

6. Ligand site (SAMHD1, 4BZC)

4BZC contains 12 copies of the ligand **T8T (dGTP-α-S)**, and seven of them touch chain A. The functional substrate site is **T8T A700**, the copy in chain A's catalytic site (contact residues include 149–150, 164, 207, 210, 215, 233–234, 311–319, 366–383 around the HD metal centre). **My best kept pocket (pocket 3, rank 1, score 0.483) covers 100% of its 17 contact residues**, so the top pocket is exactly where the nucleotide binds. The other six copies sit in the **allosteric sites** that drive tetramerisation. Each of these sites is built from residues of two or more subunits, so chain A contributes only 1–9 residues to each. The pockets that best cover them rank 11th to 28th, with scores of 0.003 or less (for example pocket 8, rank 11, covering 50% of the D800 site, and pocket 19, rank 21, covering 78% of D900). fpocket scores these real binding sites low because, in a monomer, each one is only a shallow, open half-site at what would be a subunit interface. They are also lined by charged residues that grip the phosphates, and fpocket favours enclosed, hydrophobic cavities. This is the 'missing partners' limit in action. It matters for SAMHD1 because allosteric activators would bind exactly there.

Anchor check: p53 (from lab Part 2)

In lab Part 2 the AlphaFold p53 model matched chain B of 1TUP (residues 96–289) with an **RMSD of 0.68 Å over 192 confident residues** (0.73 Å over all 194), well inside the 'very close' range. The largest differences were residue 121 (4.1 Å), in the L1 loop whose neighbour K120 grips DNA in the crystal; residues 183–187 (up to 3.7 Å), where p53 copies touch each other on the DNA in 1TUP; and the chain ends (96–97, 289). These reflect DNA and partner chains holding loops in place in the crystal, which the prediction lacks, rather than real errors. Of the Module 3 hotspots, **C176, H179, C238 and C242 are zinc ligands** (2.1–2.3 Å from the zinc), **R248 and R273 are DNA contacts** (3.4 and 2.5 Å), R249 is near DNA (7.2 Å), and R175, Y220, G245 and R282 are structural, touching neither directly. All eleven sit in the confident core (pLDDT 94–98). The three positions the Module 3 language model found least tolerated from sequence alone (176, 238, 242) are all zinc ligands, so its 'intolerant' signal tracks real structural constraint. R175H, one of the most common cancer mutations, scored only −5.97 (19.3rd percentile), and structure explains why: R175 is 5.3 Å from the zinc but binds neither zinc nor DNA, acting indirectly by destabilising the zinc region. Y220 is about 29 Å from both the zinc and the DNA, and its druggable Y220C crevice does not appear in a wild-type prediction. Reading this anchor correctly is what lets me trust the same workflow on SAMHD1 and APOE.

References

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